Are home buyers inattentive towards energy efficiency?

An econometric analysis based on a new data set created with NLP and OCR techniques

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Welcome

On this site, I present my empirical project, conducted as part of my Master's thesis in Applied Economics and Data Science. In this project, I attempt to determine whether homebuyers in Denmark value energy efficiency and specific energy-related features in properties.

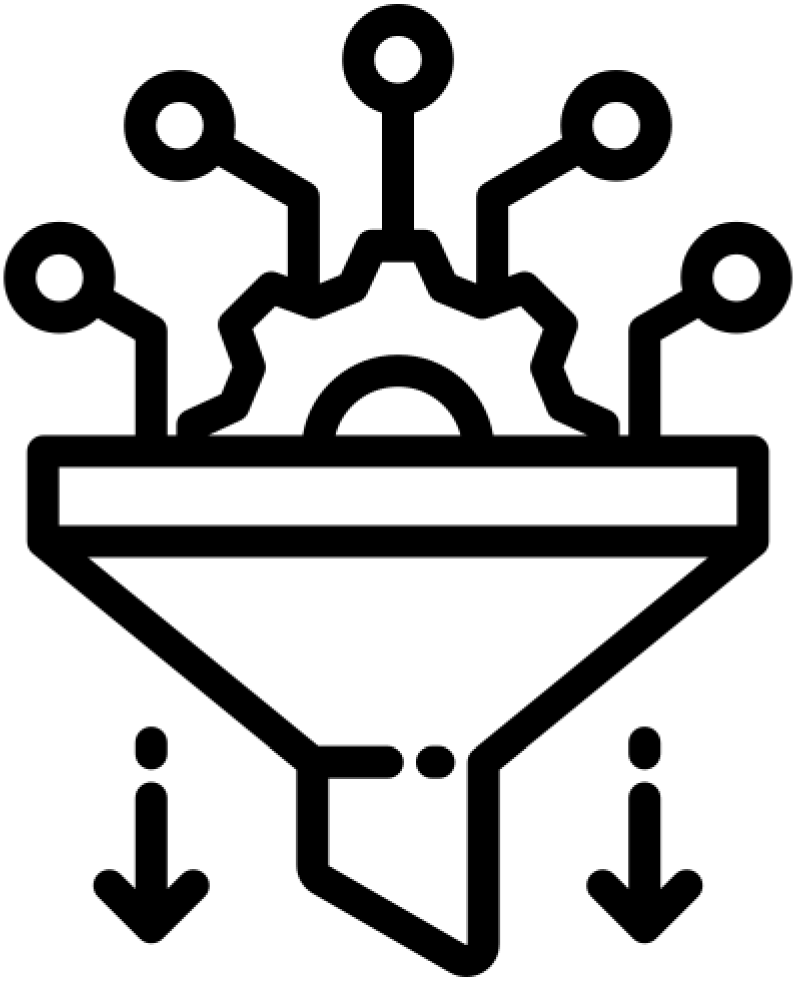
In particular, I explore whether homebuyers are willing to pay a premium for energy-efficient properties or, conversely, accept a discount for properties that are less energy-efficient. To quantify energy efficiency both the energy label or EPC label (i.e. the label stated in the Energy Performance Certificate of each property) and estimated energy consumption in kWh was utilized.

Furthermore, the impact of different energy heat sources and three energy systems—specifically, properties with heat pumps, solar panels, and solar heating—on property prices was examined.

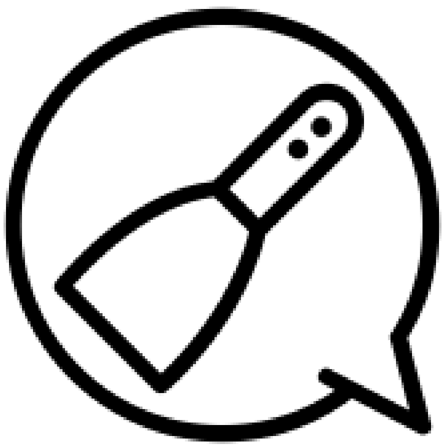
Key languages/tools used

Programming Languages and their functions

Python

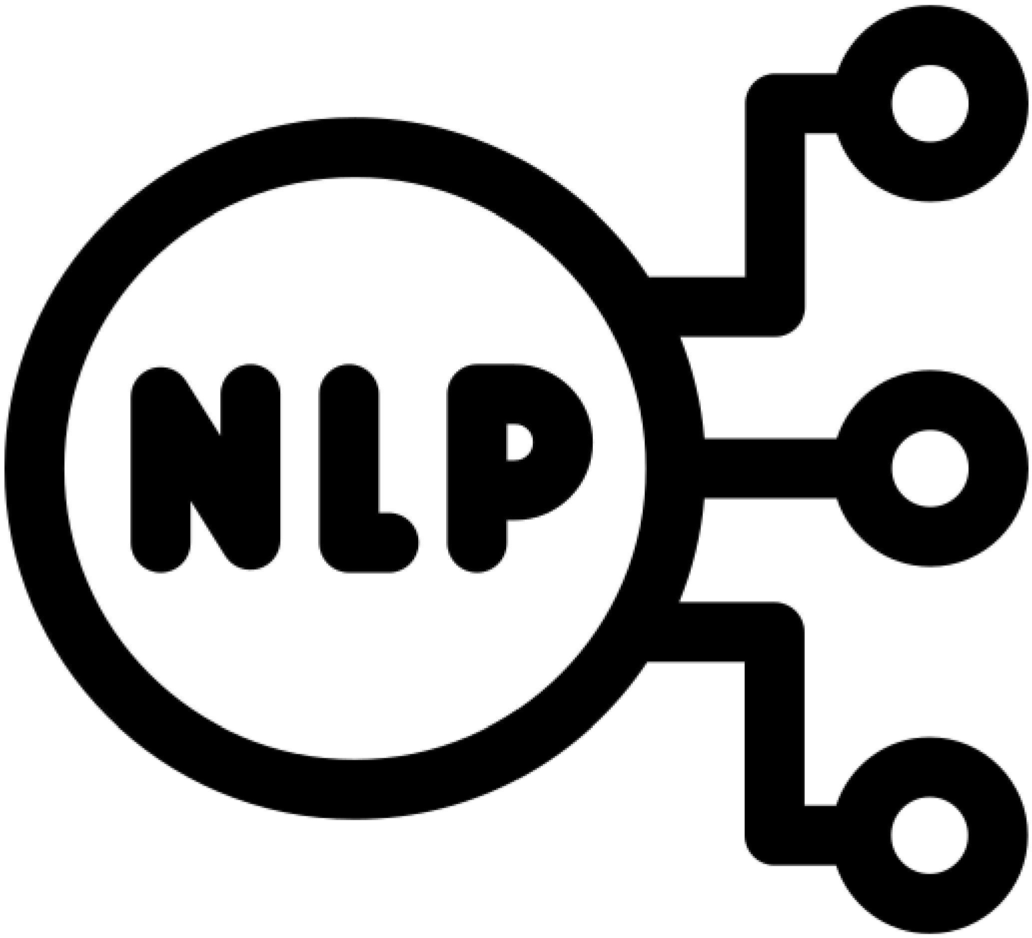


- Data cleaning and pre-p



- Web scraping

- PDF parsing (The EPC reports)



- Text classification mode

- Optical character recog

- Geographical data extrac

Data and tool related to each function

1 Introduction

In the last decades, addressing the issue of climate change has become increasingly urgent. Governments worldwide have embarked on ambitious plans to lower carbon dioxide emissions and reduce energy consumption due to this heightened awareness. The European Union has established a long-term strategy to become a climate-neutral economy with net-zero greenhouse gas emissions. With this plan, the EU aims to establish a crucial stepping stone for the world to achieve the goal of the Paris Agreement by keeping global warming below 2 degrees Celsius. This goal emphasized the urgent need for substantial economic change to reduce carbon dioxide emissions, particularly in high-emissions sectors. Among those sectors is the building industry, which is estimated to be responsible for 40% of global carbon dioxide emissions (Carlin 2022). About 70% of these emissions are generated by building operations, while construction activities contribute the remaining 30% (Carlin 2022). Regarding the European Union, the European Environment Agency (2023) published a report stating that the building sector is responsible for about 35% of energy-related emissions in the European Union for the year 2021.These emissions are primarily connected to activities like the combustion of fossil fuels within buildings, such as oil and gas boilers for heating, alongside electricity consumption for lighting, water heating, and cooling systems. In 2002, the EU introduced the Energy Performance of Buildings Directive (EPBD) with the main objective of promoting energy efficiency (EC 2002). The European Union acknowledged the importance of a new policy instrument to tackle the emissions from the building

EPBD introduced the energy performance certificates (EPC) to tackle both the information failures, i.e. the information gap on the energy efficiency of properties that exists between property owners and prospective buyers, as well as the behavioral failure, such as decision-making heuristics and biases. This initiative aims to provide property buyers with accurate, direct, and cost-free information to ensure they can make well-informed decisions that would otherwise be contained by lack of transparency. Additionally, the information that the EPC reports provide can serve as a motivator for both owners and builders to invest in energy-efficient measures since it can be hypothesized that higher property prices and rents can result from a building's improved energy performance. The Directive did not, however, compel property owners to publicly reveal the energy label, i.e., their property's energy performance, in an advert. However, a recast of the EU Directive on the energy performance of buildings was set in place in 2010. This recast imposed on Member States to make it a requirement that property owners include the energy performance indicator, i.e. the energy performance rating ranging from A to G, in any commercial media real estate advertisement (EU 2010).

Interestingly, between 2005 and 2021, total carbon dioxide emissions from the EU building sector dropped by 31% (European Enviroment Agency 2023)

Since 1997, it has been mandatory in Denmark to have an energy certification. This makes Denmark one of the first countries in the EU to implement an EPC scheme, which includes benchmarking, an energy label, and an energy performance rating in the building sector. Interestingly, even though Denmark became one of the first to implement the system, they did not observe an impact on the real estate market until 2011, i.e., one year after the recast of the EPBD (Jensen, Hansen, and Kragh 2016). The 2010 recast sparked many research studies to investigate the relationship between the energy performance of properties and property prices. The majority of the studies that investigate the EU housing market focus on the energy label. Other studies focus either on estimated or measured energy consumption. The results of these studies, however, have been mixed. Some studies identify significant price premiums associated with high energy performance labels or low energy consumption, whereas others observe no premium at all. Overall, the majority of studies identify significant price premiums, indicating a general trend toward the valuation of energy-efficient properties (Cespedes-Lopez et al. 2019). Concerning the Danish housing market, there is only a small amount of existing literature on the subject where the latest research that was found was conducted in the year 2016 (see Christensen et al. 2014; Jensen, Hansen, and Kragh 2016; and Næss-Schmidt, Heebøll, and Fredslund 2015). Furthermore, only a limited number of studies have taken into account additional information stated in the EPC reports, such as the renovation recommendations, renovations cost, and investment cost stated in the reports or the potential energy label a property can obtain from implementing recommended renovations.

This study investigates the relationship between energy performance and property prices by accounting for additional information taken from the reports. A hedonic model is applied to a unique data set of properties in Denmark from the years 2010 to 2023. Information is extracted using regular expressions, OCR, and NLP text classification techniques from over 776,106 EPC reports and then combined with scraped property characteristics and transaction data consisting of over 2.2 million observations, obtained from the website boliga.dk. Once all data is merged and cleaned, we obtain a final sample consisting of over 728,794 observations.

With this unique dataset, the contributions of this thesis are threefold. Firstly, it is the first paper to account for all residential property types in Denmark. To the best of our knowledge, all studies that investigate the Danish housing market have only focused on detached houses. Secondly, besides the energy label and energy consumption, we incorporate additional information from the EPC reports related to renovation potentials toward energy efficiency. The Idea here is that if consumers pay attention to the information in the EPC reports and are willing to pay a price premium for energy-efficient property, hence a discount for non-energy efficient property, then that premium/discount might vary depending on the renovation potential and the energy efficiency that the property can achieve from it. This might play a significant role in non-energy efficient properties. Finally, we measure the effect of tighter policy implementation for the Danish housing market on property renovations. The Danish authorities implemented tighter building regulations on January 1, 2018, which set minimum energy performance requirements for all new buildings and minimum energy efficiency requirements for renovations of existing buildings.

The thesis is structured as follows: Chapter 2: EPBD and the Danish EPC scheme introduces the implementation of the Energy Performance Certificates in Denmark and the structure of the reports. Subsequently, Chapter 3: Literature Review looks into the literature related to this paper. Chapter 4: Data chapter gives a detailed description of the data and how it was obtained. Chapter 5: Empirical Strategy outlines the theory of the hedonic model and the empirical strategy utilized in this paper. Finally, in the Chapter 6: Results, we present the results.

2 EPBD and the Danish EPC scheme

With the establishment of the Energy Performance of Buildings Directive (EPBD) in 2001, the EU constructed its main legislative tool to improve the energy performance of buildings. With this Directive, EU members were given a framework to develop regulations for the energy efficiency of buildings. The framework provided a rounded structure on matters such as defining energy performance and methodology to quantify and measure the energy efficiency of buildings (EC 2002). Thus, the EU Commission highlighted the importance of energy conservation in buildings and formulated the EPC system. From 2006 onwards, the EPC system was gradually implemented across member states, with all states mandated to have an operational system in place by the final deadline of 4 January 2009 (Arcipowska et al. 2014). Since 2002, the EU Commission has updated the Directive through the years. In 2010, the EU recast the EPBD (European Directive 2010/31/EU) by updating the previous Directive as well as improving the quality of the Energy Performance Certification scheme by implementing additional requirements regarding independent qualified and/or accredited experts authorised to perform assessments of a building's energy performance (EU 2010). Moreover, The EU Commission later expanded the scope of the recast in January 2012 by introducing a comparative methodology framework to determine the most cost-effective levels of minimum energy performance standards for both buildings and building elements. The recast of the EPBD played a vital role in enhancing transparency in the methodology of the energy performance of buildings and minimising potential information asymmetries between the property buyer and seller. Before the recast, sellers were not obligated to publicly display the energy performance of the property. Sellers could simply showcase the certificate at the time of signing the purchase agreement or rental contract. However, this changed with the recast, which required that when buildings

"are offered for sale or for rent, the energy performance indicator of the energy performance certificate of the building or the building unit, as applicable, is stated in the advertisements in commercial media" (EU 2010, 24)

The fundamental motivations of the recast was to push for more sustainable development and make critical information more accessible to potential buyers and renters such that it fosters greater demand for energy-efficient buildings. By doing so, the goal is to reduce energy consumption and greenhouse gas emissions in the housing sector. As a result, EPCs simplify the procedure of comparing similar buildings and can help boost awareness in the market towards energy efficiency, which intends to assist the EU in achieving its sustainability goals. (Arcipowska et al. 2014)

Members of the European Union have a degree of freedom to develop their measurement systems, accreditation requirements, and EPC layouts under the subsidiarity principle. Nevertheless, they are required to be in line with the EU framework policy given in the EPBD and meet the minimum methodology requirement specified in Annex 1 in the Directive. Certain key information must be displayed in the EPCs to fulfil EPBD requirements. This includes the energy performance index, which reflects the energy efficiency rating of the building, as well as reference values, such as minimum energy performance requirements, enabling comparison between different buildings. Expert recommendations are also required, where possible, and feasible home renovations that can improve the energy performance of the building are listed. Finally, the methodology estimate needs to be standardised at the national level. (EU 2010)

A report made by Arcipowska et al. (2014) gives a clear overview of the implementation of EPC across the EU and the different countries that implement an EPC scheme. From the report, it can be seen that over the course of the implementation, the majority of countries have developed a letter-based rating scheme, for instance, on a scale from A to G, where A is most efficient and G is least efficient. These ratings are determined either based on specific energy consumption values or by comparing the building's performance to reference buildings. Furthermore, the majority of countries have made it a requirement that qualified experts must make an on-site visit to the property to issue an energy performance certificate for existing buildings, that includes Denmark. The on-site requirement is one of the ways to ensure that the quality of the input data is upheld for the calculation process.

Since the year 1997, it has been mandatory in Denmark to have an energy certification. Denmark was the first EU member to construct a complete building certification scheme, which consisted of an energy performance label system ranging from labels A to G. The Danish Energy Agency projected a positive impact on market activity by promoting the sale of highly energy-efficient buildings from an early implementation. Additionally, Danish policymakers argued that the implementation of the energy performance rating for buildings would not only raise the energy efficiency standards of the building stock, but would also serve as an important motivator in driving the society's overall efforts towards reducing carbon emissions (Jensen, Hansen, and Kragh 2016). As a result, starting from 1 January 1997, every building that was listed for sale was required to have a valid energy performance rating (Jensen, Hansen, and Kragh 2016). Interestingly, Jensen, Hansen, and Kragh (2016) point out in their paper that even though Denmark became one of the first EU members to implement a scheme, there was no observable effect for 15 years. The Danish authorities implemented the 2010 EBPD recast almost two weeks after it got passed, i.e. on 1 July 2010. One year later, an effect was observed on the market. Danish real estate agents claimed that the easiest properties to sell were properties with higher energy performance. (Jensen, Hansen, and Kragh 2016)

The Danish EPC scheme is maintained by the Danish Energy Agency (DEA). They are responsible for monitoring, maintaining quality assurance, and conducting further developments on the program. In order to receive an EPC report for a given building, a property must be inspected by a licensed energy consultant, which then will determine the building's energy performance (Concerted Action EPBD 2020). The licensed energy consultant calculates the building's energy consumption and signs a label on a scale from A to G, where A indicates the highest energy standard and G the lowest. He inspects the building's quality concerning insulation, windows and doors, heating installation system, heating sources, etc. On this basis, the building's energy consumption is calculated under certain assumptions such as weather, family size, operating hours, consumption habits, etc ("Energimærkning" n.d.). In other words, the energy consumption is theoretical. However, it is set up to reflect normal household consumption, given the building's attributes. With that being said, it may not necessarily reflect the actual energy consumption, which is, of course, highly dependent on both the weather and the habits of the residence. Moreover, the EPC reports give an overview of the energy improvements that the energy consultants deem cost-effective to implement at the time of reporting.

Over the years, the energy labels in Denmark have changed multiple times. Table **?@tbl-EPC_table** depicts the conversion from all the different energy labels that the Danish EPC scheme has had to the current labelling system, as well as the current official energy thresholds for each label. To this day, the A label in Denmark is split into three subcategories: A2020, A2015 and A2010. Where A2020 represents the most energy-efficient category, and G is labelled the least efficient. The 2013 scale is the official EPC labels in Denmark today. To this day, newly constructed buildings must have an energy label A2015 or A2020 ("Hvornår Skal Bygninger Energimærkes?" 2016). As Table **?@tbl-EPC_table** shows, a house of the size 100 square meters labelled D has an energy consumption between 167 (135+3200/100) and 217 (175+4200/100) kWh per square meter per year, i.e. between 16.700 to 21.700 kWh per year. On the other hand, a property of the same size with the label A2020 has an energy consumption equal to or less than 25 kWh per square meter per year i.e., 2,500 kWh per year or less.

Table 2.1: Conversion table - EPC classification for residential buildings

2006 scale	2008V1 scale	2008V2 scale	2011 scale	2013 scale (Current scale)	Threshold limit 2013- ($kWh/m^2/year$)
-	-	-	-	A2020	≤ 25
A1	-	-	A1	A2015	$\leq 41.0 + 1000/A$
A2	A	A1,A2	A2	A2010	$\leq 71.3 + 1650/A$
B1	B	B	B	B	$\leq 95.0 + 2200/A$
B2,C1	C	C	C	C	$\leq 135 + 3200/A$
C2,D1	D	D	D	D	$\leq 175 + 4200/A$
D2,E1	E	E	E	E	$\leq 215 + 5200/A$
E2,F1	F	F	F	F	$\leq 265 + 6500/A$
F2,G1,G2	G	G	G	G	$> 265 + 6500/A$

As mentioned earlier, this thesis will analyse the period 1. July 2010 to 2023. To quantify the label effects across the whole sampling period, all labels from older labeling systems are converted into the newest system based on the conversion table. Furthermore, within the period of this study, the layout and design of the Danish EPC reports have had three major changes. This is important because not all designs convey the exact same information. For clarity, the designs of the certificates will be referred to as Design 1, Design 2, and Design 3, where Design 1 is the oldest and Design 3 is the newest. The biggest difference is between the first design and the other designs. Essentially the renovation recommendation includes the following information: the type of renovation, estimated investment cost, and energy cost saved. In all designs, the renovations are categorised into two groups: profitable renovations and other renovations. The first group represents so-called "profitable renovations," i.e., renovations that the licensed energy expert deems profitable based on estimated costs and savings. A renovation is deemed profitable if the energy savings can pay back the investment before the proposed renovation needs to be replaced again. As an example, if the proposal is to replace a circulation pump for hot water, the pump is expected to last for 15 years, and the savings proposal is considered profitable if the estimated energy savings can pay back the investment over 15 years. The second group will be referred to as the "non-profitable renovations", which represent renovation proposals that cannot repay the investment before the proposed renovation needs to be replaced again. However, these renovations are often beneficial to consider if the building is being renovated or if there are building components that need to be replaced anyway. The profitable renovations include both investment cost, energy savings in Danish krone (DKK), and energy consumption saved. For non-profitable renovations, only the energy savings in DKK and the energy consumption saved are included. In addition to this, both EPC designs 2 and 3 include the energy label that a building would get from implementing profitable renovations and the energy label it would get from implementing all renovations, i.e. both profitable and non-profitable renovations. However, design one only includes the potential energy label the property would get from all renovations. Overall, in chapter 4: Data we will go in more detail how we harmonise our data over all the designs.

3 Literature review

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Part I

Data

4 Housing data

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5 Energy performance certificates data

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6 Geographical data

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Part II

Empirical Strategy

7 Empirical Strategy

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8 Empirical Strategy

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9 Results

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10 Discussion

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11 Appendix

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